Measure the Door, Not the Exit

Five statistics for the AI-and-work debate, and who should collect them

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The problem in one paragraph

The public argument about artificial intelligence and employment runs on occupational exposure indices. Their numbers are large and they are easy to quote: about four in five U.S. workers hold an occupation in which at least a tenth of tasks could be done by a language model; about one in five holds one in which half could. None of those numbers observes an employer. An exposure index measures what a model can do to a task bundle. It cannot tell you whether any firm adopted the model, at what price, for which tasks, or what happened to headcount afterwards. Between the capability and the outcome sits the adoption decision, and no official statistic currently measures it.

This brief proposes five statistics that would. They are chosen to be cheap, to be collected by existing institutions, and to adjudicate specific disagreements rather than add another index. For each, the brief states what it settles, what it would cost, and what result would refute the displacement thesis.

Why the obvious statistics do not work

Employment levels. Sector employment moves slowly and is buffeted by the rate cycle. The information sector's 12.2 per cent decline since November 2022 is consistent with AI substitution, with higher interest rates, and with the unwinding of pandemic over-hiring. A statistic that three explanations predict equally cannot separate them.

Unemployment rates. The same problem in a smaller number. The U.S. unemployment rate is 4.2 per cent; the rate for workers aged 20 to 24 is 7.5 per cent. The relative penalty for being young has not changed: the ratio between the two rates is 2.08 today against 2.07 before the pandemic. Aggregate rates pool sectors and seniority, which is exactly where the effect is supposed to live.

Job postings. Useful and underused, but a posting is an intention. It can be withdrawn, duplicated, or written for a hire the firm never makes. Postings measure the shape of demand, not its realisation.

Exposure indices. They measure the supply of capability against the content of an occupation. They are the input to this debate, not the output of it.

The five statistics

1. Adoption intensity: model spend per employee, by task group

Definition. Firm spending on AI tools and services divided by headcount, broken out by the task group where the tool is used (writing, extraction, code, analysis, customer support). A companion measure is usage

frequency per employee, which some vendors already report to their customers.

What it settles. Whether exposure and adoption are correlated at the firm level. If firms with high exposure do not adopt, the exposure literature is describing a possibility, not a market. If they adopt but do not change hiring, the effect is in the economics of the tool, not the capability.

Who collects it. National statistical offices, through an annual business survey module. The U.S. Census Bureau's Business Trends and Outlook Survey already asks about AI use in one question; it needs spend and task detail. Tax authorities already hold the spending data at firm level and could aggregate it with a lag.

What would refute the thesis. Adoption rising while hiring in exposed task groups is flat or rising, for four consecutive quarters, with the same result in two countries.

2. Entry-level share of hires, by occupation

Definition. The share of new hires in an occupation with less than two years of relevant experience. Quarterly, by occupation, from linked employer-employee data or from job-posting composition where postings state seniority.

What it settles. Whether the adjustment runs through the entry door. My companion note on the information sector shows that the sector's contraction ran through hires and attrition rather than layoffs: hires fell 33 per cent from their October 2021 peak while layoffs stayed far below their pandemic level. That is the flow signature of substitution for junior work, and it is equally the signature of a cost-cutting freeze. The entry-level share separates them.

Who collects it. Labour ministries with administrative payroll data, or statistics offices through an establishment survey module. Commercial posting data can produce a first estimate within a quarter at low cost, with the usual caveats about representativeness.

What would refute the thesis. The entry-level share flat or rising in exposed occupations while total hiring falls: that would indicate a broad demand shock, not task substitution.

3. Separation type by tenure

Definition. Separations split into quits, layoffs, and other, by tenure band (under one year, one to three, three to ten, over ten).

What it settles. Whether the adjustment is attrition or dismissal, and at which seniority. A substitution story predicts reduced entry and normal attrition among junior staff; a recession story predicts layoffs spread across tenure.

Who collects it. JOLTS already asks separations by type but not tenure; adding a tenure band to the existing instrument is a form change, not a new survey.

What would refute the thesis. Layoffs concentrated in senior tenure bands in exposed occupations. That would point to a reorganisation, not to substitution for routine work.

4. Reinstatement rate

Definition. Of the tasks displaced from hiring in year *t*, what share reappear as hiring requirements in the same occupation within 24 and 48 months.

What it settles. The reinstatement half of the task framework. Acemoglu and Restrepo (2019) argue that technology displaces tasks and reinstates labour in new ones; the framework predicts new demand eventually. If displaced tasks never reappear, the optimistic branch of the framework is wrong for this technology.

Who collects it. Posting data, tracked longitudinally, or occupation-level administrative data. This is the most speculative of the five, because task definitions must be fixed in advance and held stable, which is expensive and political.

What would refute the thesis. Reinstatement rates at or above historical norms within four years: evidence that the adjustment is reallocation rather than loss.

5. Within-occupation wage dispersion

Definition. The ratio of the 90th to the 10th percentile of hourly wages within an occupation, annually. Reported alongside the mean, which is what most series report now.

What it settles. Whether AI compresses the return to routine cognitive work before it changes headcount. A technology that makes some tasks cheaper can widen the spread inside an occupation without moving its average: junior routine work loses value, judgment work gains it. Occupation averages cannot see this.

Who collects it. The Current Population Survey already collects wages and occupation; the estimate is a tabulation, not a new survey. Administrative wage records would be better and are held by social insurance agencies.

What would refute the thesis. Dispersion flat or narrowing in exposed occupations over four years while adoption rises.

A six-question module

The five statistics need data collection, not a new agency. A six-question annual module on an existing business survey would produce the first four, and a tabulation change would produce the fifth.

1. Did the firm use a generative AI tool in the last 12 months? (yes/no)
2. Approximate spend on AI tools and services as a share of payroll. (banded)
3. Which task groups used it? (writing, extraction, code, analysis, customer support, other)
4. Did the firm change planned hiring for any task group as a result? (increase / no change / decrease / do not know)
5. For the last 12 months, what was the share of new hires with less than two years of relevant experience? (banded)
6. For separations in the last 12 months, what share were quits, layoffs, and other? (banded)

The module is deliberately blunt. It is designed to be answered by a finance or HR manager without research support, and to be comparable across firms and countries. The bands are coarse because the first objective is the sign of the effect, not its magnitude.

What the alternative costs

The current debate spends its measurement budget on exposure indices, which are cheap to compute and impossible to falsify. A firm that adopts nothing and a firm that adopts everything score the same on an exposure index if they share an occupation mix. That is the definition of a measure that cannot answer the question it is quoted to answer.

The five statistics above cost more: a survey module, a tabulation, and administrative data access. They are also the only ones that can be wrong in a way that teaches something.

What would change my mind

I would abandon the argument in this brief if, on the statistics above, adoption rose sharply in an exposed task group while the entry-level share of hires and within-occupation wage dispersion stayed flat for eight quarters, across two countries. That pattern would mean the technology is being adopted and is changing nothing about who gets hired or what they are paid, which is a real possibility and would make the displacement literature an artefact of measurement.

I would also abandon it if reinstatement rates turn out to be normal. Then the task framework's optimistic branch is right: the technology reorganises work without reducing the demand for workers, and the policy problem is transition, not substitution.

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